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| **K L Deemed to be University** | **Department of Artificial Intelligence and Data Science -- KLBCH**<br>**Course Handout**<br>**2025-2026, Summer Term** |

| | |
|---|---|
| **Course Title** | **: DEEP LEARNING** |
| **Course Code** | **: 24AD3105** |
| **L-T-P-S Structure** | **: 3-0-2-0** |
| **Pre-requisite** | : Nil |
| **Credits** | **: 4** |
| **Course Coordinator** | **: John Bob Gali** |
| **Team of Instructors** | : Sreenath Itikela, Sreenath Itikela, John Gali, John Gali |
| **Teaching Associates** | : |
| | |
| **Syllabus :** | Introduction to deep learning, multilayer perceptron, activation functions, forward propagation, Loss functions, gradient descent, backpropagation algorithm, optimization techniques (ADAM, SGD), regularization methods, bias-variance tradeoff, weight initialization and batch Normalization. Convolutional Neural Networks (CNNs), CNN variants: LeNet, AlexNet, ZF-Net VGGNet, GoogLeNet, ResNet Object Detection Techniques: RCNN, Fast RCNN, Faster RCNN, Object Detection Techniques: YOLO. Introduction to Recurrent Neural Networks (RNNs), Vanishing and Exploding Gradients in RNNs, Loss Functions in RNNs, Long Short Term Memory (LSTM), Gated Recurrent Units (GRUs), Attention Mechanisms, Deep dream, Deep art. Autoencoders, Variational autoencoders, GAN, Applications of deep learning in computer vision, NLP, recommendation system, speech with real world examples. Implementation of deep learning models using frameworks such as TensorFlow and PyTorch, model training, evaluation, hyperparameter tuning and deployment of models. |
| **Text Books :** | [1] Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning [2] Michael Nielsen, Neural Networks and Deep Learning [3] François Chollet, Deep Learning with Python |
| **Reference Books :** | [1] Bishop, Pattern Recognition and Machine Learning [2] Hands-On Machine Learning – Aurélien Géron [3] Dive into Deep Learning – Zhang et al. |

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|---|---|
| **MOOCS :** | [1] NPTEL – Deep Learning [2] Coursera – Deep Learning Specialization [3] TensorFlow & PyTorch Official Documentation |
| **Course Objectives :** | 1. To understand fundamentals of deep learning and neural networks. 2. To analyze various deep learning architectures and training techniques. 3. To implement deep learning models for real-world applications. |

**COURSE OUTCOMES (COs):**

| CO NO | Course Outcome (CO) | PO/PSO | Blooms Taxonomy Level (BTL) |
|---|---|---|---|
| CO1 | Identify fundamentals of neural networks and deep learning applications | PO1 | 3 |
| CO2 | Analyze training mechanisms and optimization techniques | PO1 | 3 |
| CO3 | Design deep neural network architectures | PO2 | 3 |
| CO4 | Apply deep learning techniques to solve real-world problems | PO2 | 3 |
| CO5 | Design and implement deep learning models using frameworks | PO2 | 4 |

**COURSE OUTCOME INDICATORS (COIs)::**

| Outcome No. | Highest BTL | COI-2 | COI-3 | COI-4 |
|---|---|---|---|---|
| CO1 | 3 | **Btl-2**<br>Understand fundamental concept of neural networks | **Btl-3**<br>Identify applications of Deep neural networks | |
| CO2 | 3 | **Btl-2**<br>Understand the RNN and Object Detection Techniques | **Btl-3**<br>Apply CNN and its variant models to various applications | |

| Outcome No. | Highest BTL | COI-2 | COI-3 | COI-4 |
|---|---|---|---|---|
| CO3 | 3 | **Btl-2**<br>Understand the RNN Architecture | **Btl-3**<br>Apply LSTM techniques for time series and sequence data, deep dream and deep art concepts | **Btl-4**<br>Analysing attention mechanisms and Deep dream techniques |
| CO4 | 3 | **Btl-2**<br>Understand Autoencoders and image compression | **Btl-3**<br>Applu GAN and Autoregressive models | **Btl-4**<br>Analysing the performance of recommendation system, speech with real world examples |
| CO5 | 4 | | | **Btl-5**<br>Evaulating and implementing deep learning models using DL frameworks. |

**PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)**

| Po No. | Program Outcome |
|---|---|

**Lecture Course DELIVERY Plan:**

| Sess.No. | CO | COI | Topic | Book No[CH No][Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| 1 | CO1 | COI-2 | Introduction of Deep Learning | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 2 | CO1 | COI- | Perceptron - Types | T1 | Chalk,PPT,Talk | End Semester Exam,SEM- |

| Sess.No. | CO | COI | Topic | Book No[CH No][Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| | | 2 | | | | EXAM1 |
| 3 | CO1 | COI-2 | Activation functions | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 4 | CO1 | COI-2 | Feedforward Neural Networks | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 5 | CO1 | COI-2 | Loss Functions | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 6 | CO1 | COI-3 | Backpropagation | T1 | Chalk,PPT,Talk | End Semester Exam Online,Home Assignment,SEM-EXAM1 |
| 7 | CO1 | COI-3 | Optimization techniques (ADAM, SGD), Gradient Descent | T1 | Chalk,PPT,Talk | ALM,End Semester Exam,SEM-EXAM1 |
| 8 | CO1 | COI-3 | Bias Variance Tradeoff, regularization methods | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 9 | CO1 | COI-3 | Weight Initialization Methods Batch Normalization | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 10 | CO2 | COI-2 | Introduction to Convolutional Neural Networks | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |

| Sess.No. | CO | COI | Topic | Book No[CH No][Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| 11 | CO2 | COI-2 | Convolutional Neural Networks Variants | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 12 | CO2 | COI-2 | LeNet, AlexNet | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 13 | CO2 | COI-2 | VGGNet, , ZF-Net | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 14 | CO2 | COI-2 | GoogLeNet | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 15 | CO2 | COI-2 | ResNet | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 16 | CO2 | COI-3 | Introduction to Object Detection | T1 | Chalk,PPT,Talk | End Semester Exam,HA,SEM-EXAM1 |
| 17 | CO2 | COI-3 | Object Detection types | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 18 | CO2 | COI-3 | RCNN, Fast RCNN | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 19 | CO2 | COI-3 | Faster RCNN | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |

| Sess.No. | CO | COI | Topic | Book No[CH No][Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| 20 | CO2 | COI-3 | YOLO | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM1 |
| 21 | CO3 | COI-2 | Introduction to Recurrent Neural Networks | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 22 | CO3 | COI-2 | Vanishing and Exploding Gradients | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 23 | CO3 | COI-2 | Loss Functions in RNNs | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 24 | CO3 | COI-2 | Long Short Term Memory (LSTM) | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 25 | CO3 | COI-2 | Gated Recurrent Units | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 26 | CO3 | COI-3 | Problem Solving using LSTM | T1 | Chalk,PPT,Talk | ALM,End Semester Exam,SEM-EXAM2 |
| 27 | CO3 | COI-3 | Solving the vanishing gradient problem with LSTMs | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 28 | CO3 | COI-3 | Attention Mechanisms | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |

| Sess.No. | CO | COI | Topic | Book No[CH No][Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| 29 | CO3 | COI-3 | Deep Dream and Deep Art | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 30 | CO4 | COI-2 | Introduction to Autoencoders | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 31 | CO4 | COI-2 | Variational autoencoders | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 32 | CO4 | COI-2 | Generative Adversarial Networks (GANs) | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 33 | CO4 | COI-2 | Applications of deep learning in computer vision | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 34 | CO4 | COI-2 | NLP | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 35 | CO4 | COI-2 | recommendation system | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 36 | CO4 | COI-3 | speech with real world examples | T1 | Chalk,PPT,Talk | ALM,End Semester Exam,SEM-EXAM2 |
| 37 | CO4 | COI-3 | speech with real world examples | T1 | Chalk,PPT,Talk | End Semester Exam,Home Assignment,SEM-EXAM2 |

| Sess.No. | CO | COI | Topic | Book No[CH No][Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| 38 | CO4 | COI-3 | GAN in real time applications | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |
| 39 | CO4 | COI-3 | GAN in real time applications | T1 | Chalk,PPT,Talk | End Semester Exam,SEM-EXAM2 |

**Lecture Session wise Teaching – Learning Plan**

SESSION NUMBER : 1

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Introduction of Deep Learning | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Introduction of Deep Learning | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 2

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Perceptron | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Perceptron - Types | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 3

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Activation functions - Introduction | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Activation functions - Types | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 4

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Neural Networks | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Feedforward neural networks | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 5

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Loss Functions | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Loss Functions - Types | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 6

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Backpropagation | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Backpropagation - Algorithm | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 7

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Optimization techniques (ADAM, SGD) | 2 | PPT | One minute paper |
| 20 | Gradient Descent | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 8

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Bias Variance Tradeoff | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Regularization methods | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 9

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Weight Initialization Methods | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Batch Normalization | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 10

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Introduction to Convolutional Neural Networks | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Convolutional Neural Networks Layers | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 11

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Concepts of CNN | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Convolutional Neural Networks Variants | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 12

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | LeNet | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | AlexNet | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 13

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | VGGNet | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | ZF-Net | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 14

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | GoogLeNet | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | GoogLeNet | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 15

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | ResNet | 2 | PPT | Group Discussion |
| 20 | ResNet | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 16

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Introduction to Object Detection | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Object Detection - Types | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 17

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Object Detection | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Object Detection - Types | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 18

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | RCNN | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Fast RCNN | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 19

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Faster RCNN | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Faster RCNN | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 20

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | YOLO | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | YOLO | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 21

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Introduction to Recurrent Neural Networks | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Recurrent Neural Networks - Types | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 22

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Vanishing using RNN | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Exploding Gradients using RNN | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 23

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Loss Functions | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Loss Functions using RNNs | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 24

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
| --- | --- | --- | --- | --- |
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Long Short Term Memory (LSTM) | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | LSTM Advantages | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 25

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
| --- | --- | --- | --- | --- |
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Gated Recurrent Units | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | GRU Adavantages and disadvantages | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 26

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Requirement discussion for LSTM | 3 | PPT | Quiz/Test Questions |
| 20 | Problem Solving using LSTM | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 27

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Vanishing gradient reuirements with LSTM | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Vanishing gradient problem using LSTM | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 28

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Attention Mechanism | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Attention Mechanisms using GRU | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 29

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Deep Dream and Deep Art | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Deep Dream and Deep Art with example | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 30

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Introduction to Autoencoders | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Autoencoders -Types | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 31

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Variational autoencoders | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | Decoders | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 32

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Generative Adversarial Networks (GANs) | 2 | PPT | --- NOT APPLICABLE --- |
| 20 | GAN - Applications | 2 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 33

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Applications of deep learning | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Applications of deep learning in computer vision | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 34

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | NLP | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | NLP - Applications in Deep learning | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 35

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Recommendation system | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Recommendation system using deep learning | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 36

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Speech with real world examples | 3 | PPT | Case Study |
| 20 | Real world examples with CV | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 37

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | speech with real world examples | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Real world examples using AE | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 38

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | GAN | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | GAN in real time applications | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 39

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | GAN | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | GAN in real time applications | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Summary | 3 | PPT | --- NOT APPLICABLE --- |

## Tutorial Course DELIVERY Plan:

NO Delivery Plan Exists

**Tutorial Session wise Teaching – Learning Plan**

No Session Plans Exists

**Practical Course DELIVERY Plan:**

| Tutorial Session no | Topics | CO-Mapping |
|---|---|---|
| 1 | Design deep neural network model using multi-layer perceptron. | CO5 |
| 2 | Implementing the gradient descent update rule | CO5 |
| 3 | Implementing the Stochastic Gradient Descent update rule | CO5 |
| 4 | Implement a simple neural network for image classification using PyTorch | CO5 |
| 5 | Analyse the forward pass and backward pass of back propagation algorithm | CO5 |
| 6 | Build and train a ConvNet in Pytorch for a classification problem | CO5 |
| 7 | Build a deep learning model which classifies cats and dogs using CNN using Pytorch | CO5 |
| 8 | Implement Recurrent Neural networks with pytorch | CO5 |
| 9 | Implement a Long Short-Term Memory (LSTM) network using PyTorch | CO5 |

| Tutorial Session no | Topics | CO-Mapping |
| --- | --- | --- |
| 10 | Building a Simple Autoencoder on MNIST Data using Pytorch | CO5 |
| 11 | Implement Advanced Autoencoder for Feature Extraction on MNIST Data | CO5 |
| 12 | Implement Advanced Generative Adversarial Networks (GANs | CO5 |
| 13 | Implement Autoregressive Models - PixelRNN and PixelCNN | CO5 |

**Practical Session wise Teaching – Learning Plan**

SESSION NUMBER : 1

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
| --- | --- | --- | --- | --- |
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on deep neural network model using multi-layerperceptron | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 2

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on gradient descent update rule | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 1 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 3

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on Stochastic Gradient Descent update rule | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 4

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Precap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on simple neural network for image classification | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 50 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 5

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on forward pass and backward pass of back propagation algorithm | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 6

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Build and train a ConvNet in Pytorch for a classification problem | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 7

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on Building a deep learning model which classifies cats and dogs | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 8

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on implementing Recurrent Neural networks with pytorch | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 9

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on implementing a Long Short-Term Memory (LSTM) network using PyTorch | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 10

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on building a Simple Autoencoder on MNIST Data | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 11

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on Iimplementing a Advanced Autoencoder for Feature Extraction on MNIST Data | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 12

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on implementing a Advanced Generative Adversarial Networks (GANs) | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution 5 | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 13

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance/Recap | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Demo on Autoregressive Models - PixelRNN and PixelCNN | 2 | PPT | --- NOT APPLICABLE --- |
| 50 | Execution and Evalution | 4 | Talk | --- NOT APPLICABLE --- |
| 20 | Viva | 4 | Talk | --- NOT APPLICABLE --- |

**Skilling Course DELIVERY Plan:**
NO Delivery Plan Exists

**Skilling Session wise Teaching – Learning Plan**

No Session Plans Exists

**WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDEDED PROBLEM-SOLVING EXERCISES etc:**

| Week | Assignment Type | Assignment No | Topic | Details | co |
|---|---|---|---|---|---|

**COURSE TIME TABLE:**

| | Hour | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| Day | Component | | | | | | | | | |
| **Mon** | Theory | H-S1 | H-S1 | --- | --- | -- | --- | -- | --- | --- |
| | Tutorial | -- | -- | --- | --- | -- | --- | -- | --- | --- |
| | Lab | -- | -- | --- | --- | H-S1 | --- | H-S1 | --- | --- |
| | Skilling | -- | -- | --- | --- | -- | --- | -- | --- | --- |
| **Tue** | Theory | --- | --- | H-S1 | H-S1 | --- | --- | --- | -- | -- |
| | Tutorial | --- | --- | -- | -- | --- | --- | --- | -- | -- |
| | Lab | --- | --- | -- | -- | --- | --- | --- | H-S1 | H-S1 |

| | Hour | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| | Skilling | --- | --- | -- | -- | --- | --- | --- | -- | -- |
| **Wed** | Theory | H-S2 | H-S2 | --- | --- | -- | --- | -- | --- | --- |
| | Tutorial | -- | -- | --- | --- | -- | --- | -- | --- | --- |
| | Lab | -- | -- | --- | --- | H-S2 | --- | H-S2 | --- | --- |
| | Skilling | -- | -- | --- | --- | -- | --- | -- | --- | --- |
| **Thu** | Theory | --- | --- | H-S2 | H-S2 | --- | --- | --- | -- | -- |
| | Tutorial | --- | --- | -- | -- | --- | --- | --- | -- | -- |
| | Lab | --- | --- | -- | -- | --- | --- | --- | H-S2 | H-S2 |
| | Skilling | --- | --- | -- | -- | --- | --- | --- | -- | -- |
| **Fri** | Theory | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Tutorial | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Lab | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Skilling | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| **Sat** | Theory | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Tutorial | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Lab | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Skilling | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| **Sun** | Theory | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Tutorial | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Lab | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Skilling | -- | -- | -- | -- | -- | -- | -- | -- | -- |

**REMEDIAL CLASSES:**

Supplementary course handouts, including special lectures and discussions, will be planned and the schedule will be notified accordingly.

**SELF-LEARNING:**

Assignments to promote self-learning, survey of contents from multiple sources.

| S.no | Topics | CO | ALM | References/MOOCS |
|---|---|---|---|---|

**DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:**

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

| S.no | Advanced Topics, Additional Reading, Research papers and any | CO | ALM | References/MOOCS |
|---|---|---|---|---|

## EVALUATION PLAN:

| Evaluation Type | Evaluation Component | Weightage/Marks | | Assessment Dates | Duration (Minutes) | CO1 | CO2 | CO3 | CO4 | CO5 |
|---|---|---|---|---|---|---|---|---|---|---|
| End Semester Summative Evaluation Total= 40 % | Lab End Semester Exam | Weightage | 16 | | 120 | | | | | 16 |
| | | Max Marks | 50 | | | | | | | 50 |
| | End Semester Exam | Weightage | 24 | | 180 | 6 | 6 | 6 | 6 | |
| | | Max Marks | 100 | | | 25 | 25 | 25 | 25 | |
| In Semester Formative | Continuous Evaluation - Lab Exercise | Weightage | 10 | | 90 | | | | | 10 |
| | | Max Marks | 50 | | | | | | | 50 |

| Evaluation Type | Evaluation Component | Weightage/Marks | | Assessment Dates | Duration (Minutes) | CO1 | CO2 | CO3 | CO4 | CO5 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Evaluation Total= 24 %** | **Home Assignment and Textbook** | Weightage | 6 | | 50 | 1.5 | 1.5 | 1.5 | 1.5 | |
| | | Max Marks | 40 | | | 10 | 10 | 10 | 10 | |
| | **ALM** | Weightage | 8 | | 50 | 2 | 2 | 2 | 2 | |
| | | Max Marks | 40 | | | 10 | 10 | 10 | 10 | |
| **In Semester Summative Evaluation Total= 36 %** | **Lab In Semester Exam** | Weightage | 8 | | 90 | | | | | 8 |
| | | Max Marks | 50 | | | | | | | 50 |
| | **Semester in Exam-II** | Weightage | 14 | | 90 | | | 7 | 7 | |
| | | Max Marks | 50 | | | | | 25 | 25 | |
| | **Semester in Exam-I** | Weightage | 14 | | 90 | 7 | 7 | | | |
| | | Max Marks | 50 | | | 25 | 25 | | | |

**ATTENDANCE POLICY:**

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course
In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

**DETENTION POLICY :**

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

**PLAGIARISM POLICY :**

Plagiarism is treated as a serious academic offense, and the university enforces a strict zero-tolerance policy to uphold academic integrity. For a first offense, minor cases may result in a warning and reduced marks, while major cases may lead to a zero grade and mandatory academic integrity training. A second offense will result in a zero grade, formal reprimand, and academic probation, whereas a third offense may lead to suspension and a permanent record of misconduct. Repeated violations can result in expulsion, and in group work, all members are held accountable unless responsibility is clearly established.

**COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:**

| Name of Faculty | Delivery Component of Faculty | Sections of Faculty | Chamber Consultation Day (s) | Chamber Consultation Timings for each day | Chamber Consultation Room No: | Signature of Course faculty: |
|---|---|---|---|---|---|---|
| Sreenath Itikela | L | 2-MA | - | - | - | - |
| Sreenath Itikela | P | 2-MA | - | - | - | - |
| John Gali | L | 1-MA | - | - | - | - |
| John Gali | P | 1-MA,2-MA | - | - | - | - |

**GENERAL INSTRUCTIONS**

Students should come prepared for classes by reviewing the prescribed video lectures in advance and carrying the required textbook(s) or study material(s) as specified by the Course Faculty.

**NOTICES**

All course-related notices and communications will be shared through the official institutional email.

## Signature of COURSE COORDINATOR

(John Bob Gali)(10008)(AI&DS)

**Signature of Department Prof. Incharge Academics & Vetting Team Member**

Department Of AI&DS

**HEAD OF DEPARTMENT:**

**Approval from: DEAN-ACADEMICS**
(Sign with Office Seal)
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